TAMIL NADU STATE BOARD - STANDARD 11 MATHEMATICS VOLUME 1

CHAPTER 1: SETS, RELATIONS AND FUNCTIONS

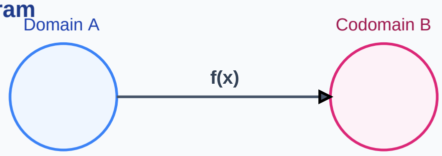
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 11 English Medium
Subject:	Mathematics Volume 1
Chapter Name:	Sets, Relations and Functions

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Function Mapping Diagram



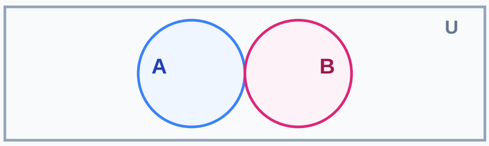
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Equivalence Relation	A relation R on a set A is equivalence if it is reflexive (aRa), symmetric (aRb => bRa), and transitive (aRb, bRc => aRc) for all a, b, c in A.
Power Set	The set of all subsets of set A, denoted by P(A). The number of elements in P(A) is 2^n where n is the cardinality of A.
Bijective Function	A function that is both injective (one-to-one) and surjective (onto), guaranteeing the existence of a unique inverse function.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

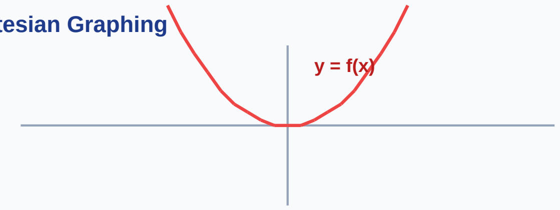
Venn Diagram: A ∩ B



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Mathematical Cartesian Graphing



Essential Formulas, Laws & Identities:

- n(A U B U C) = n(A) + n(B) + n(C) - n(A n B) - n(B n C) - n(A n C) + n(A n B n C)
- Composite Function: (g o f)(x) = g(f(x))
- Inverse Function condition: f(f^-1(y)) = y and f^-1(f(x)) = x

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Show that the relation R on Z defined by aRb iff a-b is divisible by 5 is an equivalence relation.

Step-by-step Solution:

(1) Reflexive: a-a = 0 (divisible by 5). (2) Symmetric: If a-b = 5k, then b-a = -5k (divisible by 5). (3) Transitive: If a-b = 5k and b-c = 5m, then a-c = 5(k+m) (divisible by 5). Thus, R is equivalence.

Linear Algebraic Matrix Representation

$$\begin{bmatrix} a11 & a12 & a13 \\ a21 & a22 & a23 \end{bmatrix}$$

HOTS (Higher Order Thinking Skills) Challenge

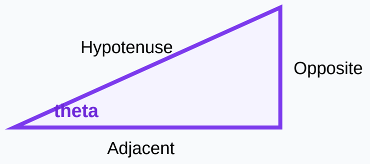
Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: If $f(x) = (x+1)/(x-1)$, show that $f(f(x)) = x$.

Analytical Solution Strategy:

$f(f(x)) = f((x+1)/(x-1)) = [((x+1)/(x-1)) + 1] / [((x+1)/(x-1)) - 1] = (x+1+x-1) / (x+1-x+1) = 2x/2 = x$.

Right-Angled Triangle Geometry



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Define equivalence class of an element.

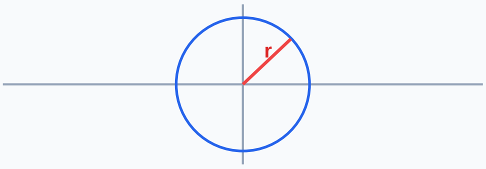
Q2 (2-Mark): If A and B have 3 and 4 elements, find the total number of relations from A to B.

Q3 (2-Mark): Let f(x) = x^2. Find the domain and range of the function.

Q4 (5-Mark): Find the inverse of f(x) = 3x - 5.

Q5 (5-Mark): Prove that (A n B)' = A' U B' using set builder notation.

Circle with Radius Vector on Plane



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Relational Algebra

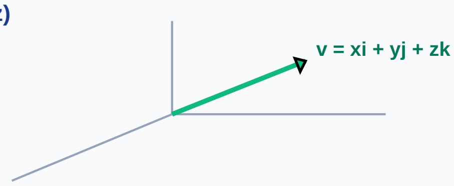
Database query optimizations use set relations and projections to retrieve indexes efficiently.

Application Case: Functional Programming

Higher-order functions in software engineering directly model composition g(f(x)) of bijective functions.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

3D Vector Projections (x, y, z)



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.

✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.

✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3, N) for the final numeric answer.

✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

[ ] Read and memorized core definitions on Page 2.

[ ] Reviewed critical formulas and relations on Page 3.

[ ] Solved worked examples on Page 4.

[ ] Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress. Good luck!

Probability Tree Outcome Diagram